

Guide to this Project

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This guide accompanies the article. It maps its numbered statements to proof locations, compares the cited and implemented proof methods, and indexes reusable classical results. The article’s Section 2 fixes the analytic framework; Sections 3 and 4 treat the torus and whole space. Source locations refer to the accompanying repository snapshot.

1 Build requirements

Use Lean 4.34.0-rc2 and the checked-in manifests, which pin Mathlib to 85e3a25e006c35636f0e53b0e9296caca2685bc0. Build in Linux:

```
lake -d verification exe cache get
lake -d verification build
```

The build checks 37 current publication interfaces. The separate declaration audit checks the transitive axioms of the results cited in this guide. For macOS 27, use a Linux VM because of the pinned runtime’s allocator issue. The repository README gives the build details.

2 Finding the theorem corresponding to the article

Each entry gives the proof file, declaration line and theorem name. The prefixes F/ and B/ mean `formalization/NSFormalization/` and `verification/Bindings/`, respectively. Names are local to the indicated file’s namespace; line numbers refer to the source tree identified above. A `def` entry is explicitly a constructed witness, not a theorem declaration. Some final proofs assemble several results in `Bindings`; these are distinct from the conformance wrappers in `verification/Tests/`.

Closed means that the complete article statement has a Lean kernel-checked proof, including every auxiliary result and its instantiation. No unproved theorem input, `sorry/admit`, or extra axiom is allowed. Only the article’s stated hypotheses and the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound` remain.

All 26 numbered statements and the numbered remark appear below and are Closed. The table follows the article’s numbering; only its two main theorems retain parenthetical titles. Qualifications record differences in scope. Historical identifiers retain *insertion* and *packet* where the article uses *gluing* and *building block*.

Article statement	Proof location and scope
Theorem 1.1	Closed. F/Source/PacketBreakdown.lean:76, theorem <code>source_breakdown</code> . Construction and same-force global nonexistence for every positive viscosity, with blowup time fixed at one.

Article statement	Proof location and scope (continued)
Proposition 2.1	Closed. B/LocalForceTheory.lean:16, theorem localForceTheory; B/LocalTheoryV2.lean:93, def localTheoryV2; F/Section4/A02/Uniqueness.lean:120, theorem velocity_unique; F/Section4/A02/Maximal.lean:157, theorem exists_maximal_of_localSolution; F/Section4/A04/ShiftedExtension.lean:247, theorem extendsBeyond_of_memForceR'; B/TorusLocalTheory.lean:483, def torusLocalTheory; B/ContinuationV3.lean:29, theorem continuationV3_holds; B/TorusLocalTheoryV2.lean:48, theorem torusLocalTheoryV2_holds; F/Section4/A04/H1Restart.lean:261, theorem h1RestartR; F/Section3/T11/H1Restart.lean:576, theorem h1RestartT. Local existence, full local regularity, maximality, uniqueness and integral continuation hold without global time-integrability or temporal support assumptions. The additional fixed-force H^1 restart interfaces retain the density force classes.
Lemma 2.2	Closed. F/Section3/T14/PackageEnergy.lean:160, theorem energy_le_work_of_packet; F/Section3/T14/PackageEnergy.lean:106, theorem work_eq_square_of_packet; F/Section4/I01/Quiet.lean:53, theorem exists_packet_quiet.
Theorem 3.1	Closed. F/Section3/T21/MainAssembly.lean:84, theorem mainStatement_holds.
Lemma 3.2	Closed. F/Section3/T13/Assembly.lean:330, theorem localizationAPI. Includes the interior-order estimate and the two endpoint identities.
Proposition 3.3	Closed. F/Section3/T15/Assembly.lean:130, theorem scalingStatement_holds.
Lemma 3.4	Closed. F/Section3/T16/Assembly.lean:483, theorem localPotential.
Lemma 3.5	Closed. F/Section3/T17/ArticleScope.lean:44, theorem correctionStatementArticle_holds; F/Section3/T17/ArticleScope.lean:70, theorem article_force_identification; F/Section3/T17/ArticleScope.lean:119, theorem article_correction_derivative_bound; F/Section3/T17/ArticleScope.lean:129, theorem article_force_derivative_bound; F/Section3/T17/ArticleScope.lean:103, theorem article_force_spatial_volume; F/Section3/T17/ArticleScope.lean:111, theorem article_force_time_length; F/Section3/T17/ArticleScope.lean:140, theorem article_correction_energy_bound; F/Section3/T17/ArticleScope.lean:155, theorem article_force_mixed_bound; F/Section3/T17/ArticleScope.lean:173, theorem article_force_sobolev_bound. Classical reference, building-block placement and the Lemma 3.4 ball; zero extension agrees on the classical slab and preserves every displayed bound.
Theorem 3.6	Closed. F/Section3/T19/FromData.lean:20, theorem periodicInsertion_from_data; B/PeriodicInsertionV2.lean:10, theorem periodicInsertionStatementV2_holds. From raw data for every prescribed coordinate ball with closure in the cube interior; all four clauses, including shrinking support in that chart, hold simultaneously.
Proposition 3.7	Closed. F/Section3/T19/Assembly.lean:43, theorem periodicDensityStatement_holds.
Proposition 3.8	Closed. F/Section3/T20/Assembly.lean:53, theorem criticalRegularityStatement_holds.
Corollary 3.9	Closed. F/Section3/T21/MainAssembly.lean:80, theorem nonDensityStatement_unconditional.

Article statement	Proof location and scope (continued)
Corollary 3.10	Closed. F/Section3/T19/Assembly.lean:47, theorem mixedRegionStatement_holds.
Corollary 3.11	Closed. F/Section3/T19/Assembly.lean:51, theorem strongClosureStatement_holds.
Proposition 3.12	Closed. F/Section3/T19/Assembly.lean:55, theorem projectionStatement_holds.
Remark 3.13	Closed. F/Section3/T18/ForceAmplitude.lean:24, theorem packetForce_ne_zero; F/Section3/T18/ForceAmplitude.lean:240, theorem forceAmplitude_lower; F/Section3/T18/ForceAmplitude.lean:288, theorem forceAmplitude_diverges; F/Section3/T18/ForceAmplitude.lean:332, theorem forceAmplitude_real_diverges; B/ForceAmplitude.lean:40, theorem forceAmplitude_from_data. The positive finite packet amplitude gives the displayed lower bound and divergence for every inserted family.
Corollary 3.14	Closed. B/BoundaryInsertionV2.lean:27, theorem boundaryInsertion_from_data; F/Section3/T22/Assembly.lean:25, theorem boundedDomainNorm.
Proposition 3.15	Closed. B/AffineVariation.lean:109, theorem affineVariationPacket. Instantiated at the selected upstream building block.
Proposition 3.16	Closed. F/Section3/T24/MultipleAssembly.lean:51, theorem multipleRegionsStatement_holds; F/Section3/T24/MultipleOmegaAssembly.lean:59, theorem multipleRegionsOmegaStatement_holds; B/MultipleRegionsV2.lean:167, theorem multipleRegionsStatementV2_holds. Both torus and arbitrary nonempty bounded open domain branches, with homogeneous no-slip on the latter.
Proposition 3.17	Closed. F/Section3/T24/ConservativeAssembly.lean:45, theorem conservativeForcing; F/Section3/T24/ConservativeOmega.lean:127, theorem conservativeForcingOmega. Periodic potentials on the torus. On any nonempty bounded open domain with homogeneous no-slip, the potential need only be smooth near the closed spatial slab during the lifespan; no boundary regularity is required.
Theorem 4.1	Closed. B/MainThresholds.lean:85, theorem mainThresholds. Density: B/DensityFromInsertion.lean:30, theorem breakdownDenseR_of_subcritical; converse: B/MainThresholds.lean:34, theorem mainThresholds_nonDensity.
Theorem 4.2	Closed. B/InsertionFromData.lean:150, theorem wholeSpaceInsertion_holds. Any prescribed nonempty open set, with one family and the given reference solution.
Proposition 4.3	Closed. F/Section4/R43/Universal.lean:193, theorem universal_of_memForceR.
Proposition 4.4	Closed. F/Section4/R44/Prop44.lean:24, theorem main. Finite-horizon $L_t^2 H_x^{-1/2}$ estimate in the inhomogeneous norm, with zero initial velocity.
Corollary 4.5	Closed. B/ForceClasses.lean:46, theorem forceClasses.
Proposition 4.6	Closed. B/CompletedDensity.lean:14, theorem completedDensity.
Theorem 4.7	Closed. B/GridObservations.lean:47, theorem gridObservations_choose. Observation times are strictly below the terminal time.

Upstream building block. Under vendor/NavierStokesAndEuler/NavierStokes/, R3ActualCandidate.lean:18 proves NavierStokes.R3CompactCandidate.selected_compact_candidate; R3FiniteEnergyComparison.lean:37 proves NavierStokes.ComparatorBridge.compact_candidate_excludes_global_solution. In ComparatorR3Theorem.lean:36, NavierStokes.ComparatorBridge.navier_stokes_breakdown_R3 combines the candidate,

uniqueness comparison and viscosity rescaling. `Source/PacketBreakdown.lean` applies positive-viscosity uniqueness to the selected construction and proves the complete statement of Theorem 1.1 for the same force.

The unnumbered analytic inputs are indexed with literature sources and proof locations in Section 4.

3 Formalized scope and proof methods

Section 2 records the full formalized statements. The comparisons below describe the original proof routes retained by the density theorems and the broader results now available.

Local theory and continuation (Proposition 2.1). Proposition 2.1 is formalized for locally smooth forcing on both domains, without global time-integrability or compact temporal support assumptions. The theorem `localForceTheory` provides local existence with full Sobolev regularity and pressure recovery, maximal development, uniqueness and the squared- H^2 integral continuation criterion. Whole-space forcing is represented by smooth future Sobolev paths; periodic forcing is represented by smooth extensions on each finite closed future slab. The whole-space proof localizes the force in time, while the periodic proof uses force bounds on a finite time window. Force agreement and uniqueness transfer the constructed solution and its continuation to the original force.

The downstream density proofs continue to invoke the original H^7 local construction on $\mathbb{R}^3$ and H^3 construction on the torus for their density force classes. Their continuation argument applies the high-order energy estimate and Grönwall to obtain

$$\sup_{t < S} \|u(t)\|_{H^m} \leq \left(\|a\|_{H^m} + \int_0^S \|f(t)\|_{H^m} dt \right) \exp \left(C_{m,\nu} \int_0^S \|u(t)\|_{H^2}^2 dt \right) < \infty.$$

Taking $m = 7$ or 3 gives a bounded set of restart data. A uniform positive restart time for the fixed force on a compact time window then allows uniqueness to glue the solution across S . Tao’s Theorems 5.1 and 5.4 [1] start with a contraction in the H^1 mild-solution space and propagate higher regularity on the same interval; Theorem 5.4(iv), p. 53, treats smooth data in all Sobolev orders.

Fixed-force H^1 -uniform restart and endpoint continuation are also formally proved for the density force classes, using the enstrophy inequality and a uniform ODE barrier. The original H^7/H^3 route is recorded in `F/Section4/A04/RestartFixedForce.lean:41`, theorem `restartFixedForce_of_memForceR`, `F/Section4/A04/ShiftedExtension.lean:247`, theorem `extendsBeyond_of_memForceR` and `F/Section3/T11/ExistenceInputH3.lean:456`, theorem `extendsBeyondH3`; the registered H^1 conclusions appear in the correspondence row for Proposition 2.1.

Bounded domains (Propositions 3.16 and 3.17). Both bounded-domain statements are formalized for every nonempty bounded open set Ω . Proposition 3.16 uses the article’s finite superposition of building blocks supported in disjoint interior balls. Proposition 3.17 uses the vanishing force–velocity pairing and the energy identity from rest. The required integration-by-parts identity is proved on arbitrary bounded open sets; the potential need only be smooth near $[0, T) \times \bar{\Omega}$.

Fractional localization (Lemma 3.2). The article invokes Taylor’s coordinate-cover and partition-of-unity construction of Sobolev spaces, with interpolation [2, Chapter 4, Sections 2–3]. A fixed chart cutoff then gives the required bounded local-to-global map. The formal proof establishes the estimate directly for the torus: it identifies homogeneous fractional norms with double-integral seminorms on $\mathbb{R}^3$ and on the torus, compares the periodized kernel with the Euclidean kernel, and bounds the remaining lattice tail by a multiple of $\|z\|_2^2$. The fixed distance from the support ball to

the chart boundary makes the constant uniform as the support shrinks. The endpoint orders 0 and 1 are treated separately. The kernel comparison is `F/Section3/T13/KernelComparison.lean:394`, theorem `iTorus_periodize_le`; the resulting estimate, including both endpoints, is `F/Section3/T13/Assembly.lean:330`, theorem `localizationAPI`.

Critical regularity and non-density. The article invokes the forced critical fixed-point theory [8, Theorem 2.3.1 and Lemma 2.3.2, pp. 47–49], with the inhomogeneous heat estimate (Theorem 2.2.5, p. 44) for the finite-horizon $L_t^2 H_x^{-1/2}$ case. The density proofs retain their original route through the formalized critical energy inequalities, a smallness bootstrap and the high-order continuation argument above. See the proof locations for Propositions 3.8, 4.3 and 4.4 in Section 2. The open-ball argument yielding non-density is unchanged; only the analytic proof route differs.

4 Classical results available for future reference

The following classical results have formal proofs in this project and may provide starting points for later formalizations. The entries give actual proof locations using the prefixes from Section 2. Each entry gives a literature locator and identifies any specialization or derivation needed to reach the formal version. Unless stated otherwise, the spatial domain is $\mathbb{R}^3$ and vector fields take values in $\mathbb{R}^3$.

1. **Sobolev embedding into bounded functions.** For smooth fields with square-integrable derivatives through order two, the code proves the pointwise and essential-supremum H^2 bounds: `F/Section4/A03/BoundedRepresentative.lean:165`, theorem `enorm_le_jetENorm` and `F/Section4/A03/BoundedRepresentative.lean:182`, theorem `eLpNormTop_le_jetENorm`. These use the derivative-based norm `jetENorm`. *Source:* Tao [3, Appendix A, (A.13), draft p. 336], with $d = 3$, $s = 2$. Apply the scalar estimate componentwise and use Plancherel to pass to the equivalent derivative norm; the literal constant is not claimed to coincide with the code’s constant.
2. **Sobolev algebra and tame product estimates.** At integer orders $m \geq 2$, scalar products satisfy the H^m algebra estimate, and scalar–vector products satisfy the tame estimate with an H^2 low norm: `F/Section4/A03/ScalarTameProduct.lean:308`, theorem `algebraProductScalar` and `F/Section4/A03/VectorTameProduct.lean:273`, theorem `tameProductVector`. *Source:* Tao [3, Appendix A, Lemma A.8, (A.17)–(A.18), draft p. 338]. Formula (A.18) gives the algebra estimate; (A.17) combined with (A.13) gives the H^2 low norm in dimension three. The vector version follows componentwise.
3. **Integral representations of fractional Sobolev norms.** For $0 < s < 1$, the double-integral seminorm equals an explicit constant times the squared homogeneous Fourier norm: `F/Section3/T13/WholeSpaceIdentity.lean:509`, theorem `wholeSpace_identity` for smooth compactly supported fields, and `F/Section3/T13/TorusIdentity.lean:1174`, theorem `torus_identity` for smooth periodic fields, with the mean removed in the homogeneous norm. *Sources:* Di Nezza–Palatucci–Valdinoci [4, Proposition 3.4, arXiv v3, pp. 16–17] for $\mathbb{R}^3$; Roncal–Stinga [5, arXiv v3, Theorem 1.5, (1.6)–(1.8), p. 3; Fourier definition (1.1), p. 1] for the torus. The periodic energy identity is a consequence of their operator formula: set $\sigma = 2s$, pair with the field and symmetrize the integral. Parseval then gives the squared Fourier norm. Rescale their 2π -periodic torus to period one and adjust Fourier constants; the constant mode contributes zero.
4. **Smooth approximation in Sobolev and Bochner spaces.** For every real s , compactly supported smooth vector fields are dense in the implemented H^s datum space: `F/Section4/B01/Spatial.lean:134`, theorem `spatialApprox`. For $1 \leq q < \infty$, Sobolev-valued $L^q(0, \infty)$ paths can be approximated by finite sums $\sum_j \phi_j(t) A_j$, with $\phi_j \in C_c^\infty(0, \infty)$: `F/Section4/B01/Temporal.lean:170`, theorem `temporalApprox`. The temporal theorem alone does

not make the spatial data A_j smooth; the spatial approximation supplies that separate step. *Sources:* Taylor [2, Chapter 4, Section 1, Exercise 1, author-chapter p. 5] gives Schwartz density for every real s . Multiplying a Schwartz approximant by $\chi(x/R)$ gives convergence in the Schwartz topology and hence in H^s , yielding compact support; this last step is a deduction, not the exercise’s literal statement. Hunter [6, Appendix 6.A, Proposition 6.29, p. 200] gives the finite separated-sum approximation on $(0, T)$. Truncating the integrable q -th power tail first extends it to $(0, \infty)$ for $q < \infty$.

5. **A local vector potential for a divergence-free field.** On a ball, a smooth time-dependent divergence-free field has a smooth vector potential given by the radial integral: `F/Section3/T16/BallPotential.lean:324`, theorem `exists_potential_on_ball`. *Source:* Petersen [7, Section 7.1, Lemma 7.1.2, pp. 127–128, and Lemma 7.1.5, p. 129]. Apply the homotopy formula to the closed 2-form $\iota_v(dx_1 \wedge dx_2 \wedge dx_3)$ and the radial contraction $x \mapsto x_0 + \rho(x - x_0)$. Translating its primitive into vector notation gives the code’s radial integral. Smooth dependence on time follows by differentiation under the integral over $0 \leq \rho \leq 1$. This specializes the Poincaré lemma to a ball, rather than arbitrary domains.
6. **Classical Navier–Stokes local theory and continuation.** Periodic local existence is proved in `F/Section3/T11/ExistenceInputH3.lean:376`, theorem `exists_periodicLocalSolution_unconditional`. Whole-space velocity uniqueness and pressure uniqueness up to a time-dependent constant are in `F/Section4/A02/Uniqueness.lean:120`, theorem `velocity_unique` and `F/Section4/A02/Uniqueness.lean:244`, theorem `pressure_gauge`. The continuation theorems are listed in Section 3. These results use the project’s smooth initial-data and force classes. *Sources:* Tao [1, Theorems 5.1(ii)–(iv), p. 48, and 5.4(ii)–(iv), pp. 52–53] for local existence, uniqueness and smoothness; Lemma 4.1(i), (40), pp. 44–45, for the whole-space pressure gauge; Corollary 5.2, p. 50, for periodic maximal development and its H^1 blowup alternative. These are the classical local-theory inputs. The precise squared- H^2 continuation test and fixed-force H^7/H^3 restart used here are derived by the high-order estimate and Grönwall in Section 3; they are not quoted as literal conclusions of Tao’s theorems.

References

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